

Quantum Mechanics I

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office hours: Monday 7pm in M.A.T.H. 110

Syllabus

Course structure

15 weeks, two 100-minute lectures per week. For each week, 6 problems (2+2+2 for three levels of difficulty), a list of suggested reading (including original and experimental sources), and a 1-page¹ summary sheet are provided. The midterm exam contributes 30%, the final exam 70%. The exam questions will be variants of the weekly problems.

Course content (tentative)

Wk	Theme	lectures: keywords
I	systems, states, measurement	$\aleph_I$: the need for a non-classical description of physical systems (listing by Dirac); the spin/polarization experiment as the generic example; summary of vector spaces and complex numbers; Dirac notation and how (dual) vectors, inner products, and outer products are represented; polarization of a photon as a “classical” example of a two-level, quantum system; $\beth_I$: relation between states and vectors; spin states in Dirac notation, and how the observation of the spin/polarization measurement is described in this framework; postulates of quantum mechanics including the operator-measurement correspondence (matrix representation, eigenvalues); illustration with spin operators;
II	dynamics, symmetry	$\aleph_{II}$: discretized amplitude hopping; continuum limit; Schrödinger equation; wave function as amplitude for “acts as if”; position and momentum representations. $\beth_{II}$: unitary transformation; Hermitian generator; translation, rotation, time; conservation as $[A, U] = 0$; stationary states; eigenvalue problem.
III	harmonic oscillator	$\aleph_{III}$: algebraic solution; $a, a^\dagger$; ladder; zero point; representation change as coordinate change. $\beth_{III}$: coherent states; quadratures; phase space; the state that follows a classical trajectory, and why it is the exception. First promise redeemed in lecture XXVI.

¹Named after Paulo Bedaque who envisioned the perfect presentation in a 1-slide format to me.

Wk	Theme	lectures: keywords
IV	angular momentum I	$\aleph_{IV}$: rotation generators; $\mathfrak{su}(2)$ algebra; ladder operators; spectrum from the algebra alone. $\beth_{IV}$: matrix representations; spin- $\frac{1}{2}$ and the qubit identified; 2π rotation and the sign.
V	angular momentum II	$\aleph_V$: orbital angular momentum; spherical harmonics; parity. $\beth_V$: rotation matrices; Wigner d ; irreducible tensors; Wigner–Eckart.
VI	central forces	$\aleph_{VI}$: radial equation; centrifugal barrier; bound states; degeneracy. $\beth_{VI}$: hydrogen; scaling; accidental degeneracy and its symmetry origin.
VII	approximation	$\aleph_{VII}$: time-independent perturbation theory, non-degenerate and degenerate; variational principle; Hellmann–Feynman. $\beth_{VII}$: time-dependent perturbation theory; Fermi golden rule; adiabatic theorem; Berry phase.
VIII	magnetic fields	$\aleph_{VIII}$: minimal coupling; gauge; Landau levels. $\beth_{VIII}$: Zeeman (now with perturbation theory available); Aharonov–Bohm; geometric phase revisited.
IX	two-level dynamics	$\aleph_{IX}$: spin precession; Larmor; Rabi oscillation; rotating frame. $\beth_{IX}$: pulses as gates; Bloch-sphere trajectories; NMR as the original quantum control.
X	coupling of angular momenta	$\aleph_X$: addition; Clebsch–Gordan; singlet and triplet. $\beth_X$: fine and hyperfine structure; the 21 cm line; atomic clocks.
XI	mixed states	$\aleph_{XI}$: density matrix; purity; Bloch ball; reduced state; partial trace. $\beth_{XI}$: von Neumann’s measurement chain; where the cut is placed; decoherence and the preferred basis; Kraus maps and the Lindblad equation in outline.
XII	correlations	$\aleph_{XII}$: entanglement; Schmidt decomposition; monogamy; no-cloning. $\beth_{XII}$: Bell inequality; CHSH; loopholes; what locality and realism each cost; teleportation.
XIII	transformation theory	$\aleph_{XIII}$: representation as a choice of basis; continuous spectra; transformation functions $\langle x p \rangle$; unitary equivalence; Schrödinger, Heisenberg and interaction pictures as one content in three dresses. $\beth_{XIII}$: quantum conditions; Poisson bracket to commutator; canonical quantization and where it is ambiguous; Ehrenfest’s theorem; coherent states revisited; in what sense a limit is classical.
XIV	symmetry and conservation	$\aleph_{XIV}$: symmetry groups; generators and Lie algebra; translation, rotation, Galilean boost, and the conserved quantity belonging to each; degeneracy as a group statement; Runge–Lenz and the hydrogen degeneracy of lecture XII explained. $\beth_{XIV}$: Wigner’s theorem, symmetries are unitary or antiunitary; parity; time reversal and its antiunitarity; Kramers degeneracy; permutation symmetry stated as a postulate and deliberately not developed.
XV	action and path integral	$\aleph_{XV}$: Dirac’s 1933 paper; Feynman’s sum over paths; the propagator; the free particle done completely; stationary phase and the emergence of the classical trajectory.

Wk	Theme	lectures: keywords
		$\beth_{XV}$: the action principle; Noether and Schwinger; symmetry to conservation law inside the path integral; Aharonov–Bohm as a phase around a loop; what the advanced course opens with.

Scriptum

[N_I] The essence of quantum behavior

A description in which objects carry their properties of which they can be asked at any time by pertinent measurements anytime is not how nature works. There are, however, objects which always *answer* certain questions with a definite answer, and we want to call those objects “large”. In contrast, it was found very early that there is the other kind, the “small” system which does reply in a seemingly random way. For the latter, we want to devise a quantitative description. An all encompassing example is the behavior of light of extremely low intensity. The sought-after description must account for:

- Indivisibility,
- polarization,
- interference,
- superposition, and
- indeterminacy.

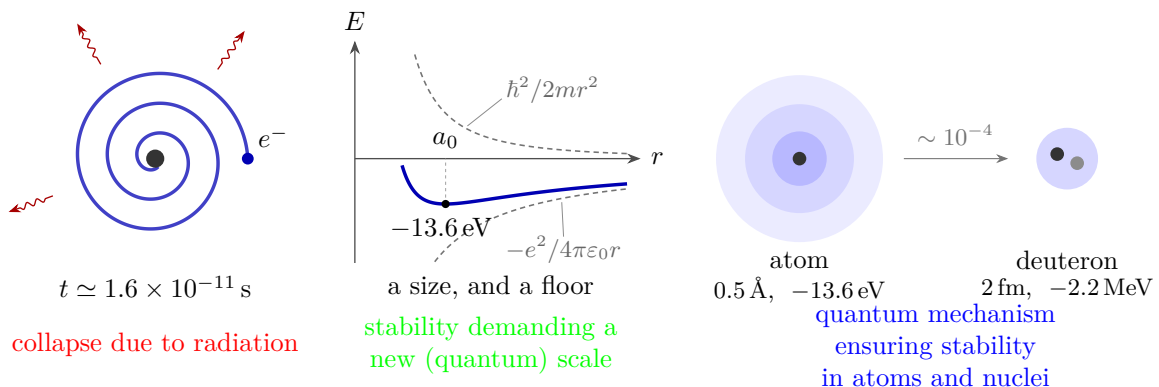
And, an explanation of Rutherford’s scattering (1911) which goes beyond the “unstable” one suggesting the atom as a planetary system of electrons orbiting a heavy nucleus by means of electromagnetic attraction.

Larmor tells us that any accelerated charge radiates away power, and a classical lifetime estimate would predict the electrons to spiral onto the nucleus in a very short time.

Dimensional analysis gives an even farther-reaching indication: there must be something at work which the classical constants m , $e^2/4\pi\epsilon_0$ and c cannot quantify: They can be combined to yield

$$r_e = \frac{e^2}{4\pi\epsilon_0 mc^2} = 2.8 \text{ fm},$$

while all other length scales are even smaller than this four-orders-of-magnitude-too-small one. In other words, there is no classical length the atomic size Rutherford was probing. [Estimate the order of magnitude and unit of a new constant of nature that could set the atomic scale $a_0 \approx 0.529 \text{ \AA}$.]



In turn, once a this novel mechanism introduces the new length scale, the stability of the atomic nucleus can/could be a consequence of the same mechanism.

But I want to start with a tangible example which should embody the key idea which represents the fundamental shift in philosophy that quantum mechanics introduces:

Classical physics: a system has properties, and measuring reveals them.

Quantum physics: a system has a state; a measurement is a question; some questions are incompatible, and answering one erases the answer to another.

It remains to specify (mathematically):

- How the state is represented? [vector]
- How is the state measured, in general, manipulated? [expectation values and operators]
- How is its evolution in time predicted? [Action principle $\rightarrow$ Schrödinger equation, &c.]

measurements on two-state systems

Let us begin by putting on Stern and Gerlach's shoes, and send Atoms into an inhomogeneous magnetic field.

[SIM] [stern-gerlach.html](#)

More generally, this asks what is being detected if a beam of particles, presumably all in the same state, is sent through an appropriate version of a Stern-Gerlach apparatus? First, let us run the experiment such that we can trace each particle, silver atom or photon or ..., individually. We find

$$\sigma(\hat{n}) = \pm 1 \quad \text{for every filter orientation } \hat{n}, \quad \text{never } 0, \quad \text{never } \cos \theta.$$

$[\sigma(\hat{n})]$: the reading of the apparatus whose axis points along the unit vector $\hat{n}$

and a spatial splitting of the beam; the classically-expected band is not found.

What happens if the apparatus is rotated? Nothing, at all, for we would still detect only ± 1 for each particle. Now, let us add another Stern-Gerlach apparatus in sequence, with its axis oriented along a different direction. The idea is to allow only one of the two states exiting the first filter to enter the second. This is what the above simulation does when the oven emits particles with a defined "orientation". Note that although we call it orientation, we do not yet know what we are dealing with except that "it" apparently dictates whether our atom ends up in the "up" or "down" path of the Stern-Gerlach apparatus.

You should now find what the two guys found long ago, namely not the

$$\underbrace{\sigma_{\text{cl}} = \hat{n} \cdot \hat{m} \in [-1, 1]}_{\text{expected, continuous}} \quad \text{but the discrete } \sigma = \pm 1 \quad \text{with the special case } \sigma|_{\hat{n}=\hat{m}} = +1.$$

$[\hat{m}]$: the preparation (polarizer) axis, $\hat{n}$: the analysis (analyzer) axis, both unit 3-vectors in ordinary space.]

Discuss the most general arrangement: polarizer oriented in $\hat{m}$ and analyzer oriented in $\hat{n}$ direction. For a single particle, the outcome is ± 1 , for all (!) orientations of oven and filter (do not confuse the behind-the-filter blocking with the actual measurement outcome).

Considering an ensemble of identically prepared particles along $\hat{m}$, we observe an average outcome

$$\langle \sigma \rangle = \hat{n} \cdot \hat{m} = \cos \theta, \quad \theta = \angle(\hat{m}, \hat{n}),$$

$[\langle \cdot \rangle]$: arithmetic mean of single-particle outcomes] which seems to measure the component of the preparation axis $\hat{m}$ along the analysis axis $\hat{n}$.

Let us now cast this in the language of probabilities: “How probable is it to get $\sigma = +1$ or $\sigma = -1$ for given ' $\hat{m}$ and $\hat{n}$ ': [$P_{\pm}$: probability that the analyzer along $\hat{n}$ reads $\sigma = \pm 1$, for a particle prepared along $\hat{m}$. They sum to one because no third reading ever occurs.]

$$P_+ + P_- = 1, \quad P_+ - P_- = \cos \theta \quad \implies \quad P_{\pm} = \frac{1 \pm \cos \theta}{2}, \quad P_+ = \cos^2 \frac{\theta}{2}, \quad P_- = \sin^2 \frac{\theta}{2},$$

$$\theta = 0 : (1, 0); \quad \theta = \frac{\pi}{2} : \left(\frac{1}{2}, \frac{1}{2}\right); \quad \theta = \pi : (0, 1).$$

Let us proceed with a setup designed to shed light on whether or not the measurements erase information encoded in the initial preparation along $\hat{z}$. With the sequence

$$\hat{z} \longrightarrow \hat{x} \longrightarrow \hat{z} \quad \text{we find} \quad \langle \sigma_z \rangle = 0, \quad ,$$

[$\sigma_z \equiv \sigma(\hat{z})$, and likewise σ_x, σ_y : the reading with the apparatus along a coordinate axis. The arrows are successive settings of one and the same apparatus, in time order. Susskind §1.4.] and thus, if we allow only the $+\hat{z}$ channel to pass through $\hat{x}$, the right $\hat{z}$ result $\langle \sigma_z \rangle = 0$ implies equal probabilities for the $+\hat{z}$ and $-\hat{z}$ outcomes.

You can analyze this with

[SIM] [stern-gerlach-three.html](#)

which you (sh/c)ould parametrize such that it reproduces the original 2-filter experiment.

Having seen all this in a context (atoms) where a different outcome was expected, recall that the observed behavior is what is classically accepted when dealing with photons and their polarization. The latter is described by the three sets of orthogonal polarization states: vertical/horizontal, diagonal/anti-diagonal, and left-/right-circular, which correspond to the three ways the atoms leave the Stern-Gerlach apparatus along three mutually orthogonal axes (magnetic fields along $\hat{x}, \hat{y}, \hat{z}$). The following pairs represent the corresponding polarization directions of the electric/magnetic field of the photon:

$$\{\hat{v}, \hat{h}\}, \quad \{\hat{d}, \hat{a}\}, \quad \{\hat{r}, \hat{l}\},$$

[$\hat{v}, \hat{h}, \hat{d}, \hat{a}, \hat{r}, \hat{l}$: the six analyzer settings vertical, horizontal, diagonal, anti-diagonal, right-circular, left-circular.] and if a photon is polarized along any one of these, the probability to find it in any state of the other two pairs is $P = \frac{1}{2}$. If the angle between initial and analyzing polarizer is θ , the probability to find the photon in the analyzing polarizer is given by Malus' law which Stern and Gerlach found for their atomic beams, too:

$$P_+ = \cos^2 \theta \quad \text{and} \quad \langle \sigma \rangle = \cos 2\theta.$$

These observations are now to be explained. They suffice to guess/postulate the formal structure of QM. First, let us get to the stage of knowledge from where we can make such a conjecture: what are complex numbers, and vector spaces (bra, kets, inner products, operators, and bases)?

complex numbers

[Susskind §1.8.]

$$z = x + iy, \quad x, y \in \mathbb{R}, \quad i^2 = -1, \quad z = r e^{i\theta} = r (\cos \theta + i \sin \theta).$$

[$r = \sqrt{z^* z} \geq 0$ is called modulus, and θ argument. Two real numbers represent one complex number.] Addition is easier in components, multiplication in polar form:

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2), \quad (r_1 e^{i\theta_1})(r_2 e^{i\theta_2}) = r_1 r_2 e^{i(\theta_1 + \theta_2)}.$$

Every z has a unique conjugate/dual z^* , with z^*z real and non-negative:

$$z = x + iy = r e^{i\theta} \iff z^* = x - iy = r e^{-i\theta} , \quad z^*z = r^2 \geq 0 .$$

We denote those numbers with $r = 1$,

$$z^*z = 1 , \quad z = e^{i\theta} = \cos \theta + i \sin \theta ,$$

phase factors or simply phases.

vector spaces

The measurement outcomes discussed above suggest the following, profound switch:

$$\text{classical system} \leftrightarrow \text{set} \quad \Leftrightarrow \quad \text{quantum system} \leftrightarrow \text{vector}$$

Let us denote these state vectors “kets”, and use the notation:

$$|\text{set of numbers which characterize the state}\rangle \stackrel{\text{e.g.}}{=} |\text{position, mass, (hair) color, (eye) color, \&c.}\rangle .$$

If vectors, those objects must satisfy the axioms of a vector space:

- (i) $|A\rangle + |B\rangle = |C\rangle$
- (ii) $|A\rangle + |B\rangle = |B\rangle + |A\rangle$
- (iii) $\{ |A\rangle + |B\rangle \} + |C\rangle = |A\rangle + \{ |B\rangle + |C\rangle \}$
- (iv) $\exists! 0 : |A\rangle + 0 = |A\rangle$
- (v) $\exists! -|A\rangle : |A\rangle + (-|A\rangle) = 0$
- (vi) $|zA\rangle = z|A\rangle = |B\rangle$
- (vii) $z\{ |A\rangle + |B\rangle \} = z|A\rangle + z|B\rangle , \quad \{ z + w \} |A\rangle = z|A\rangle + w|A\rangle$

Two-dimensional state vectors can be represented concretely as columns of complex numbers:

$$|A\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} , \quad \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} + \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} \alpha_1 + \beta_1 \\ \alpha_2 + \beta_2 \end{pmatrix} , \quad z \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} z\alpha_1 \\ z\alpha_2 \end{pmatrix} .$$

$[\alpha_i, \beta_i \in \mathbb{C}$: the components of $|A\rangle$ and $|B\rangle$ with respect to a basis that has been fixed beforehand while the abstract ket refers to the state itself without reference to any basis.]

To every “ket” belongs a “bra” with an *antilinear* correspondence:

$$|A\rangle \longleftrightarrow \langle A| , \quad |A\rangle + |B\rangle \longleftrightarrow \langle A| + \langle B| , \quad z|A\rangle \longleftrightarrow \langle A| z^* ,$$

$$|A\rangle = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_N \end{pmatrix} \iff \langle A| = (\alpha_1^* \quad \cdots \quad \alpha_N^*) .$$

The inner product is always bra times ket, and returns a complex number:

$$\begin{aligned} \langle C| \{ |A\rangle + |B\rangle \} &= \langle C|A\rangle + \langle C|B\rangle , & \langle B|A\rangle &= \langle A|B\rangle^* , \\ \langle B|A\rangle &= \sum_i \beta_i^* \alpha_i , & \langle A|A\rangle &= 1 \text{ (normalized)} , & \langle B|A\rangle &= 0 \text{ (orthogonal)} . \end{aligned}$$

The dimension of the space is the largest number of mutually orthogonal vectors in it; an orthonormal basis $|i\rangle$, $i = 1 \dots N$, then resolves any ket, the components being its inner products with the basis vectors: $[\delta_{ij}$: the Kronecker symbol, 1 for $i = j$ and 0 otherwise — here it is precisely the statement that the basis is orthonormal. Susskind §1.9.5, Eqs. 1.3–1.5.]

$$\langle j|i\rangle = \delta_{ij} , \quad |A\rangle = \sum_i \alpha_i |i\rangle , \quad \alpha_j = \langle j|A\rangle , \quad |A\rangle = \sum_i |i\rangle \langle i|A\rangle .$$

[$\sqsupset_I$] Mapping physics onto mathematics

Now let us map physics onto math: the state ket-vector identification (Dirac and row/column notation). Specifically, spin states: all there is to know, there is to measure, are the components along a given direction in space, and since any such direction in 3 dimensions is specified by two numbers, the state space is two-dimensional. Express the measurement outcomes for the xyz directions using Dirac notation $|\uparrow_x\rangle, |\downarrow_x\rangle, |\uparrow_y\rangle, |\downarrow_y\rangle, |\uparrow_z\rangle, |\downarrow_z\rangle$ and phrase that in terms of three pairs of mutually orthogonal vectors whose projections on one another are identical. As a consequence: the vector space is complex and two-dimensional.

[R_I] Reading list

- [1] O. Stern, “A Way towards the Experimental Examination of Spatial Quantisation in a Magnetic Field”, trans. by E. E. B. Campbell and B. Friedrich, *Zeitschrift für Physik D: Atoms, Molecules and Clusters* **10**, 114–116 (1988) doi:10.1007/BF01384842; trans. of “Ein Weg zur experimentellen Prüfung der Richtungsquantelung im Magnetfeld”, *Zeitschrift für Physik* **7**, 249–253 (1921) doi:10.1007/BF01332793.
- [2] W. Gerlach and O. Stern, *The Stern-Gerlach Experiment*, trans. by M. Bauer, (2023) arXiv:2301.11343 [physics.hist-ph]; trans. of “Der experimentelle Nachweis der Richtungsquantelung im Magnetfeld”, *Zeitschrift für Physik* **9**, 349–352 (1922) doi:10.1007/BF01326983.
- [3] G. E. Uhlenbeck and S. Goudsmit, “Spinning Electrons and the Structure of Spectra”, *Nature* **117**, 264–265 (1926) doi:10.1038/117264a0.
- [4] R. P. Feynman, F. L. Vernon, and R. W. Hellwarth, “Geometrical Representation of the Schrödinger Equation for Solving Maser Problems”, *Journal of Applied Physics* **28**, 49–52 (1957) doi:10.1063/1.1722572.
- [5] L. Susskind and A. Friedman, *Quantum Mechanics: The Theoretical Minimum* (Basic Books, 2014), chapters 1–3 [pedagogical].
- [6] G. Baym, *Lectures on Quantum Mechanics* (W. A. Benjamin, 1969), chapter 1 up to “Angular Momentum” [pedagogical].

[P_I] Problem list

Level 1

- P 1. Three Stern-Gerlach filters (SG_i).** A beam leaves the oven with no preparation axis. It passes SG_1 along $\hat{z}$, of which only the $\sigma = +1$ channel continues; then SG_2 along $\hat{n}$ at an angle θ to $\hat{z}$, again keeping only $\sigma = +1$; then SG_3 along $\hat{z}$.
- (a) Applying $P_{\pm}(\theta) = \cos^2 \frac{\theta}{2}, \sin^2 \frac{\theta}{2}$ once at each analyser, give the fraction of the oven output reaching each of the two SG_3 channels.
 - (b) Evaluate at $\theta = 0, \pi/2, \pi$. At which θ is the $\sigma_z = -1$ count largest, and what is it?
 - (c) Remove SG_2 and give the same two numbers.
 - (d) Suppose every atom carried a fixed label λ and each analyser passed or stopped it according to λ alone. Transmission would then be intersection of sets. Show $T_{123} = T_1 \cap T_2 \cap T_3 \subseteq T_1 \cap T_3$, so inserting SG_2 cannot raise any count. Compare with (b) and (c).

[SIM] `stern-gerlach-three.html` runs the chain; check your (a) against the counter before you trust either.

P 2. Measurements on single photons and on ensembles. A polarizer and an analyser have their transmission axes at θ_{lab} .

- (a) For a beam, $I/I_0 = \cos^2 \theta_{\text{lab}}$; for one photon, $P = \cos^2 \theta_{\text{lab}}$. The same function is being used twice. State precisely what each of the two equations asserts, and what is being counted in each.
- (b) N photons are sent one at a time. The transmitted number K is binomial with $p = \cos^2 \theta_{\text{lab}}$. Give $\langle K \rangle$ and $\text{Var } K$.
- (c) Show $\sqrt{\text{Var } K} / \langle K \rangle = \sqrt{(1-p)/(Np)}$. At $\theta_{\text{lab}} = 45^\circ$, how large must N be before the measured transmission matches Malus to one percent?
- (d) Attenuating a laser until on average one photon per second arrives does *not* make it a one-photon source. Say why, and name a measurement that separates the two.

Level 2

P 3. The three bases. In $\mathbb{C}^2$ fix an orthonormal pair $|z+\rangle, |z-\rangle$ and define

$$|x\pm\rangle = \frac{1}{\sqrt{2}}(|z+\rangle \pm |z-\rangle), \quad |y\pm\rangle = \frac{1}{\sqrt{2}}(|z+\rangle \pm i|z-\rangle).$$

- (a) Verify that each of the three pairs is orthonormal.
- (b) Compute $|\langle a|b\rangle|^2$ for every a taken from one pair and b from another. Show that all twelve numbers equal $\frac{1}{2}$.
- (c) The photon bases $\{|H\rangle, |V\rangle\}$, $\{|D\rangle, |A\rangle\}$, $\{|R\rangle, |L\rangle\}$ obey the same relations. Write $|D\rangle, |A\rangle, |R\rangle, |L\rangle$ in terms of $|H\rangle, |V\rangle$ and repeat (b) in that language.
- (d) On the sphere a basis is an axis, and $|\langle a|b\rangle|^2 = \cos^2 \frac{\theta}{2}$ with θ the angle between axes. Show $|\langle a|b\rangle|^2 = \frac{1}{2}$ holds exactly when the axes are perpendicular, and conclude that $\mathbb{C}^2$ admits at most three such bases.

P 4. discreteness from an anti-commutator. Let $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ be the Pauli matrices and $\hat{n}$ a real unit vector with polar angles (θ, ϕ) .

- (a) Write $\sigma_n = \hat{n} \cdot \vec{\sigma}$ as a 2×2 matrix.
- (b) Using $\sigma_i \sigma_j + \sigma_j \sigma_i = 2\delta_{ij}\mathbf{1}$, show $\sigma_n^2 = \mathbf{1}$ for every $\hat{n}$. Deduce that the only possible readings are ± 1 — without solving an eigenvalue problem.
- (c) Now solve for the eigenvectors and show, up to an overall phase,

$$|\hat{n}+\rangle = \cos \frac{\theta}{2} |z+\rangle + e^{i\phi} \sin \frac{\theta}{2} |z-\rangle, \quad |\hat{n}-\rangle = -\sin \frac{\theta}{2} |z+\rangle + e^{i\phi} \cos \frac{\theta}{2} |z-\rangle.$$

- (d) Compute $|\langle \hat{n}+ | z+\rangle|^2$ and recover $\cos^2 \frac{\theta}{2}$.
- (e) In the state $|\hat{n}+\rangle$, compute $\langle \sigma_n \rangle$ and show it equals $\hat{n} \cdot \hat{m}$.

[Susskind Lecture 3, Exercises 3.3 and 3.4, worked in his §3.7.]

Level 3

P 5. The Bloch sphere.

- (a) Show that $|\psi\rangle$ and $e^{i\gamma}|\psi\rangle$ give identical probabilities for every measurement, so the physical state is the ray and not the vector.

- (b) A normalised vector in $\mathbb{C}^2$ carries four real numbers; normalisation and the global phase remove two. Show that $|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle$ with $\theta \in [0, \pi]$, $\phi \in [0, 2\pi)$ covers every ray exactly once. Mind the poles.
- (c) Show that $\langle \vec{\sigma} \rangle = \hat{n}$ in the state $|\hat{n}+\rangle$: the Bloch vector is the expectation value of $\vec{\sigma}$.
- (d) Show that orthogonal states are antipodal, and mark the six states of P 3 on a sphere.
- (e) A polarization ellipse returns to itself after a laboratory rotation of π , while the state vector returns only after 2π and collects a minus sign at π . Explain how both are true at once, and connect it to the 2π rotation of a spin- $\frac{1}{2}$.
- (f) Why is the set of rays in $\mathbb{C}^2$ called $\mathbb{CP}^1$, and why is $\mathbb{CP}^1 \cong S^2$? Build the stereographic map explicitly.

P 6. Labels as an alternative model. A concrete classical model: each atom leaves the oven carrying a unit vector $\vec{\lambda}$, isotropically distributed; an analyser along $\hat{n}$ reports $\sigma = \text{sign}(\vec{\lambda} \cdot \hat{n})$; a filter keeps the atoms it reports +1 for and discards the rest, doing nothing else to $\vec{\lambda}$.

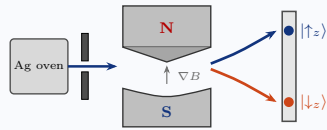
- (a) For the unprepared oven beam, show $P_{\pm} = \frac{1}{2}$ for every $\hat{n}$ and $\langle \sigma \rangle = 0$. The model survives this test — which is why the first experiment alone decides nothing.
- (b) Prepare by keeping the + channel along $\hat{m}$, so $\vec{\lambda} \cdot \hat{m} > 0$, and then measure along $\hat{n}$ at angle θ . Show that the two hemispheres overlap in a lune and hence

$$P_+ = 1 - \frac{\theta}{\pi}, \quad \langle \sigma \rangle_{\text{model}} = 1 - \frac{2\theta}{\pi}.$$

- (c) Compare with the measured $\langle \sigma \rangle = \cos \theta$. At which θ do the two agree exactly? Evaluate both at $\theta = \pi/3$, and find the θ at which they differ most.
- (d) Apply the model to the chain of P 1 and show that it gives *zero* in the $\sigma_z = -1$ channel for every θ . Identify the single assumption responsible, and state what the quantum account does in its place.

Quantum Mechanics I – Week I: 2-state systems and the Bloch Sphere

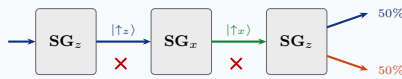
2- state atoms: Spin-(1/2)



Measurement outcomes

- classical: $\vec{\mu} \cdot \hat{n} \in [-1, 1]$, continuous
- quantum: $\sigma(\hat{n}) = \pm 1$, discrete and $\hat{n}$ independent

Sequential filtering: incompatibility



State erasure: $\hat{x}$ wipes out the prior $\hat{z}$

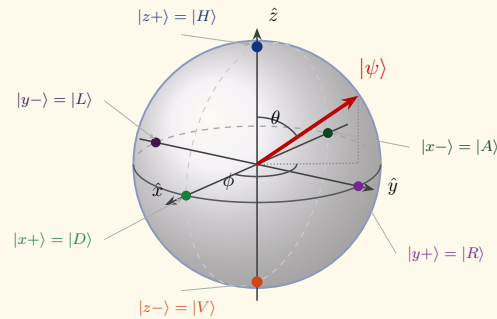
Without SG_x no counts in the lower channel. Adding SG_x counts as many clicks in the lower channel as in the upper channel.

$$T_{zz}^{\downarrow/\uparrow} = \frac{1}{4} \neq 0 \Rightarrow \langle \sigma_z \rangle = 0 \quad \text{vs.} \quad T_{zz}^{\downarrow} = 0$$

$$p_{\pm}(\theta) = \cos^2 \frac{\theta}{2}, \sin^2 \frac{\theta}{2} \quad \langle \sigma \rangle = \hat{n} \cdot \hat{m} = \cos \theta$$

$\theta = \angle(\hat{n}, \hat{m})$: angle between preparing and analyzing axes.

2-state systems: Bloch sphere



$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle$$

$$\{|z\pm\rangle\} = \{|H\rangle, |V\rangle\} \quad \{|x\pm\rangle\} = \{|D\rangle, |A\rangle\} \quad \{|y\pm\rangle\} = \{|R\rangle, |L\rangle\}$$

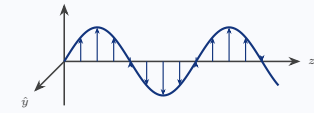
the two states are	angle apart	$ \langle a b\rangle ^2$	SG reads
the same state	0°	1	always +
partners in one basis	180°	0	never
from two bases	90°	1/2	50/50

$$|\langle \hat{n} s | \hat{m} s' \rangle|^2 = \frac{1}{2} (1 + s s' \hat{n} \cdot \hat{m})$$

$$s, s' = \pm 1, \quad \hat{n} \cdot \hat{m} = \cos \theta_{\text{Bloch}} = \cos 2\theta_{\text{lab}} \Rightarrow \cos^2 \frac{\theta}{2}, \sin^2 \frac{\theta}{2}$$

Orthogonal states are antipodal; a basis is an axis; axes are perpendicular.

2-state light: polarization

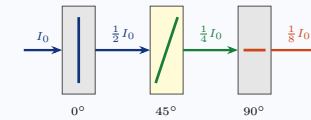


$$\vec{E}(z, t) = \text{Re}[\vec{E}_0 e^{i(kz - \omega t)}]$$

$$\vec{E}_0 = E_0 (\cos \alpha \hat{x} + e^{i\delta} \sin \alpha \hat{y})$$

basis	α	δ	$\vec{E}_0/E_0$
$ H\rangle, V\rangle$	0, $\frac{\pi}{2}$	–	$\hat{x}; \hat{y}$
$ D\rangle, A\rangle$	$\frac{\pi}{4}$	0, π	$\frac{1}{\sqrt{2}}(\hat{x} \pm \hat{y})$
$ R\rangle, L\rangle$	$\frac{\pi}{4}$	$\pm \frac{\pi}{2}$	$\frac{1}{\sqrt{2}}(\hat{x} \pm i \hat{y})$

The three-filter experiment



Single photon clicks

- the detector registers indivisible clicks $\hbar\omega$
- Malus' law governs the individual probability:

$$p(\text{click}) = |\langle \text{filter} | \psi \rangle|^2 = \cos^2 \theta_{\text{lab}}$$

- additional filter raises a count:

$$T_{0^\circ \rightarrow 45^\circ \rightarrow 90^\circ} = \frac{1}{8} \neq 0 \quad \text{vs.} \quad T_{0^\circ \rightarrow 90^\circ} = 0$$

1. HILBERT SPACE $\mathbb{C}^2$

$$|\psi\rangle = c_0 |0\rangle + c_1 |1\rangle, \quad |c_0|^2 + |c_1|^2 = 1$$

$$\text{Dual bra: } \langle \psi| = c_0^* \langle 0| + c_1^* \langle 1|$$

$$\text{Inner product: } \langle \phi | \psi \rangle = \phi_0^* \psi_0 + \phi_1^* \psi_1 \stackrel{!}{=} \langle \psi | \phi \rangle^*$$

2. OBSERVABLES AS OPERATORS

$$A = A^\dagger = \sum_i a_i |i\rangle \langle i|, \quad a_i \in \mathbb{R}$$

$$\text{Eigenvalues} = \text{outcomes } (a_i = \pm 1)$$

$$\text{Spin: } A = \vec{\sigma} \cdot \hat{n}$$

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

3. BORN RULE & PROJECTORS

$$p_i = |\langle i | \psi \rangle|^2 = \langle \psi | i \rangle \langle i | \psi \rangle = \langle \psi | P_i | \psi \rangle$$

$$\text{Projectors: } P_i = |i\rangle \langle i|, \quad P_i^2 = P_i = P_i^\dagger$$

$$\text{Identity: } \sum_i P_i = \mathbf{1}$$

$$\text{Expectation: } \langle A \rangle = \langle \psi | A | \psi \rangle = \sum_i a_i p_i$$

4. MEASUREMENT COLLAPSE

$$|\psi\rangle \xrightarrow{\text{outcome } i} \frac{P_i |\psi\rangle}{\|P_i |\psi\rangle\|} = |i\rangle$$

Post-measurement state is eigenstate $|i\rangle$

Measurement prepares the state